

DevFusion 4.0 : The Developers Hackathon



RULES HERE 

Choose any 1 Problem statement and start doing.

Problem Statement 1: Smart Campus Management Platform

Difficulty

Medium → Advanced

Domain

EdTech • SaaS • Productivity

Background

Most colleges still manage announcements, attendance, events, placements, student clubs, notices, assignments, and faculty interactions using multiple disconnected systems or even WhatsApp groups.

Build a modern Smart Campus Platform where students, faculty, coordinators, and administrators can interact through one centralized platform.

The application should be production-ready with proper authentication, role-based permissions, notifications, dashboards, analytics, and responsive design.

Objective

Develop a complete full-stack web application that enables colleges to manage:

- Student Portal
- Faculty Portal
- Event Management
- Attendance
- Placement Notices
- Club Activities

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- Assignment Submission
 - Announcements
 - Notifications
 - Admin Controls
-

Mandatory Tech Stack

Frontend

- React / Next.js
- TypeScript
- Tailwind CSS
- Responsive UI

Backend

- Node.js
- Express / Next.js API
- MongoDB / PostgreSQL

Authentication

- Google OAuth
- Email Authentication
- JWT / Session Authentication

Deployment

- Vercel
 - Railway
 - Render
 - AWS
 - Docker (Bonus)
-

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Authentication Requirements

Users must be able to

Sign Up

Using

- Google Login
 - Email + Password
-

Login

Support

- Google OAuth
 - Email Login
-

Forgot Password

Implement

- Email verification
 - OTP
OR
 - Password Reset Link
-

Email Verification

New users must verify email before accessing the dashboard.

Session Management

Support

- Secure Cookies
OR

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- JWT Tokens
-

Logout

Invalidate session securely.

Protected Routes

Unauthenticated users cannot access

- Dashboard
 - Settings
 - Profile
 - Events
 - Attendance
-

User Roles

There should be 4 roles

Student

Can

- View dashboard
- Edit profile
- Register events
- Submit assignments
- View attendance
- Placement updates
- Notifications
- Club memberships

Cannot

- Delete users
- Create notices

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- Manage attendance
-

Faculty

Can

- Create assignments
- Upload study material
- Take attendance
- Publish notices
- Review submissions

Cannot

- Delete college data
 - Manage admins
-

Coordinator

Can

- Manage events
 - Club registrations
 - Student approvals
 - Announcements
-

Admin

Full Access

Can

- Manage users
- Delete users
- Assign roles
- Analytics
- Settings
- Permissions

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- Events
- Attendance
- Placements
- Reports

Landing Page

Include

Hero Section

Features

Testimonials

Statistics

FAQ

Footer

Responsive Navigation

Dark Mode

Animations

Loading Screens

SEO

Dashboard

Every role gets a separate dashboard.

Example

Student Dashboard

Upcoming Classes

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Today's Attendance

Assignments

Events

Placement Updates

Calendar

Notifications

Quick Actions

Recent Activity

Faculty Dashboard

Classes

Attendance

Assignments

Student Count

Recent Submissions

Performance Analytics

Admin Dashboard

Total Students

Faculty

Departments

Events

Attendance %

Assignment Stats

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Placement Statistics

Charts

System Logs

Student Profile

Fields

Profile Picture

Name

Email

Phone

Roll Number

Department

Semester

Skills

LinkedIn

GitHub

Resume Upload

Bio

Attendance Module

Faculty can

Create Attendance Session

Generate Attendance

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Students can

View Attendance %

History

Subject-wise Analytics

Monthly Reports

Assignment Module

Faculty

Upload Assignment

Deadline

Attachments

Rubrics

Students

Upload Solution

PDF

ZIP

GitHub Link

Submission History

Late Submission Status

Faculty

Review

Marks

Feedback



Event Management

Admin/Coordinator

Create Event

Banner

Description

Venue

Registration Deadline

Seats

Speakers

QR Code

Students

Register

Cancel Registration

View Ticket

Download QR Pass

Placement Module

Companies

Job Roles

Eligibility

CTC

Deadline



Apply Button

Application Status

Resume Upload

Notifications

Real-time Notifications

Examples

Assignment Due

Attendance Marked

Event Reminder

Placement Open

System Alerts

Search

Global Search

Search

Students

Faculty

Events

Assignments

Placements

Analytics



Charts

Monthly Attendance

Department Performance

Assignment Completion

Placement Statistics

Event Participation

Admin Panel

Manage

Users

Departments

Courses

Events

Assignments

Attendance

Announcements

Reports

Logs

Permissions

Settings

Profile

Password



Theme

Notification Preferences

Connected Accounts

Delete Account

Privacy Settings

UI/UX Requirements

Must include

- Responsive Design
 - Dark & Light Mode
 - Loading Skeletons
 - Empty States
 - Error Pages
 - Success Screens
 - Toast Notifications
 - Beautiful Forms
 - Smooth Animations
 - Mobile Friendly
 - Accessibility (keyboard navigation, contrast, semantic HTML)
-

Security Requirements

- Password hashing (bcrypt/Argon2)
- Input validation
- Rate limiting
- CSRF protection (if applicable)
- XSS protection
- Secure cookies
- Environment variables
- File upload validation
- Authorization middleware
- Server-side validation for every protected action

- Audit logging for sensitive admin actions
-

Database Entities

Minimum collections/tables

- Users
 - Roles
 - Departments
 - Attendance
 - Assignments
 - Assignment Submissions
 - Events
 - Event Registrations
 - Notifications
 - Placements
 - Applications
 - Settings
 - Activity Logs
-

Bonus Features

- AI chatbot for campus FAQs
- QR attendance scanner
- Live chat between students and faculty
- Calendar sync
- PWA support
- Offline mode
- Multi-language support
- Face recognition attendance
- AI assignment plagiarism detection
- Email reminders
- Push notifications
- WebSockets for live updates
- Admin audit logs
- CSV/Excel export
- Dockerized deployment

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- CI/CD pipeline
 - API documentation (Swagger/OpenAPI)
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Judging Criteria

Criteria	Weight
Functionality	30%
UI/UX Design	20%
Code Quality & Architecture	15%
Authentication & Security	10%
Database Design	10%
Performance & Scalability	5%
Innovation	5%
Deployment & Documentation	5%

Expected Deliverables

- Source code (GitHub)
- Live deployed application
- README with setup instructions
- API documentation
- Database schema/ER diagram
- Demo video (3–5 minutes)
- Test credentials
- Environment variable template
- License file
- Architecture diagram